Conduct and Testing For Conduct

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Grad IO

The problem

Conduct Testing in Industrial Organization

Foundational Empirical IO Question: How do we observe data on price and quantity and infer which model of firm behavior generated those outcomes?

- Early work: Porter (1983), Bresnahan (1982,1987)
- Subsequent work defined the "menu" approach: Nevo (1998, 2001), Villas-Boas (2007)
- Recent revival of "internalization" parameters: Miller and Weinberg (2017), Crawford, Lee, Whinston, and Yurukoglu (2017), Pakes (2017)
- Parallel work by: Duarte, Magnolfi, Sølvsten, Sullivan (QE 2024): which test is best (RV) and when instruments are too weak to test. Magnolfi, Quint, Sullivan, Waldfogel (2022): should we test or estimate?
- Applications of our test: Starc and Wollmann (2022), Scuderi and Roussille (2025), others?
- Today: the classic approaches; the supply conditional moment restriction (CMR) every test is built on; the Backus–Conlon–Sinkinson scalar test ("BCS 1.0") and the full-CMR test ("BCS 2.0"); the common-ownership application.

Is conduct testable? Berry and Haile (2014): yes.

Conduct Testing in Industrial Organization

- Absent additional restrictions, we cannot generally look at data on (P, Q) and decide whether or not collusion is taking place.
  - You say we started colluding at date t, I say we received a correlated shock to mc.
- We can make progress in two ways: (1) parametric restrictions on marginal costs; (2) exclusion restrictions on supply.
  - Most of the literature focuses on (1): a functional form for marginal cost in observed cost shifters plus an unobserved shock.
  - In principle (2) is possible if we have instruments that shift demand for products but not supply. (These are much easier to come up with than "supply shifters"). (This is the point Berry Haile 2014 make).

A famous plot (Bresnahan 87)

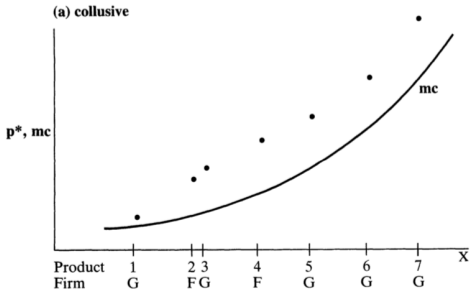


Figure 2(a)

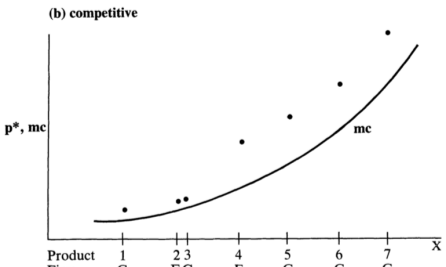


Figure 2(b)

Bresnahan (1980/1982) recognized this problem: we need “rotations of demand”.

Conduct Testing in Pictures (Berry Haile 2014)

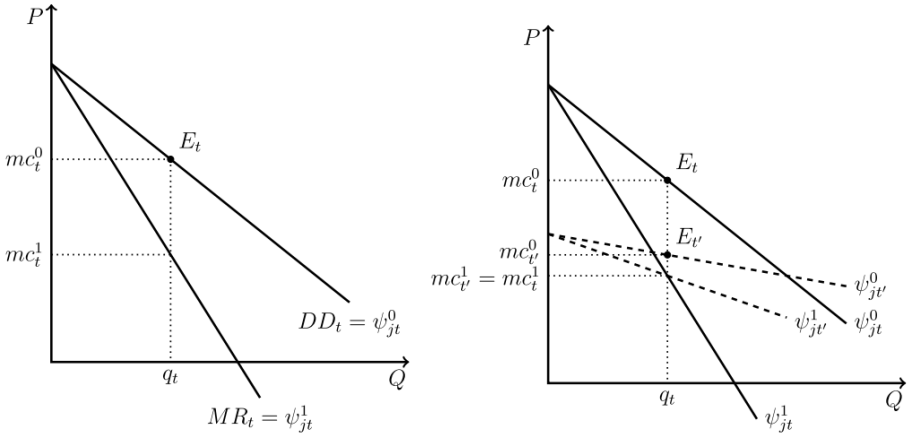


Figure 2(ab) from Berry and Haile (2014), Example 1.

Conduct Is a Markup: Profit Weights in the Bertrand FOC

Firm f puts weight κ_{fg} on rival g's profits ($\kappa_{ff} = 1$):

$$\max_{\{p_j\}_{j \in \mathcal{J}_f}} \sum_{j \in \mathcal{J}_f} (p_j - c_j)\, q_j(\mathbf{p}) + \sum_{g \neq f} \kappa_{fg} \sum_{k \in \mathcal{J}_g} (p_k - c_k)\, q_k(\mathbf{p}) \quad \Rightarrow \quad (\mathcal{H}(\kappa) \odot \Delta)(\mathbf{p} - \mathbf{mc}) = \mathbf{q},$$

with Δ the demand-derivative matrix of the pricing lecture and $\mathcal{H}(\kappa)_{(j,k)} = \kappa_{fg}$ for $j \in \mathcal{J}_f$, $k \in \mathcal{J}_g$.

- A conduct model m (an $\mathcal{H}$ matrix, or Cournot, or anything else) maps demand into a markup vector, $\boldsymbol{\eta}^m \equiv \mathbf{p} - \mathbf{mc} = \left(\mathcal{H}(\kappa_m) \odot \Delta\right)^{-1} \mathbf{q}$.
- Mergers change 0's to 1's in $\mathcal{H}$. Partial coordination is $\kappa_{fg} \in (0, 1)$.
- $\kappa_{fg} > 0$ is not necessarily evidence of malfeasance, just a deviation from static Bertrand pricing (Cournot, dynamics, common ownership, ...).

What is the Common Ownership Hypothesis?

- Economics starts from the premise that firms maximize profits.
  - Friedman (1953): natural selection of firms and billiards players.
  - Or, firms answer to investors, maximize shareholder value.
  - We generally think of these objectives as aligned (possibly subject to managers' agency frictions).
- So what do investors want?
  - Some (diversified) investors may hold stakes in you *and your competitor*. These are called "common owners."
  - Common owners are harmed when managers engage in business stealing.
- As a theory of firm behavior in joint ventures this is an old idea. The recent innovation is to extend this approach to passive or institutional investors

Profit Weights from Portfolios (BCS 2020 P&P)

Most models of common ownership (implicitly) assume:

Investor portfolios Investor s is entitled to a share β_{fs} of firm f's profits: $V_s = \sum_f \beta_{fs}\pi_f$.

Manager agency (Rotemberg 1984) Managers maximize a γ-weighted average of investor payoffs, $U_f = \sum_s \gamma_{fs}V_s$ (s could be the manager herself).

Rearranging,

$$U_f \propto \pi_f + \sum_{g \neq f} \underbrace{\left(\frac{\sum_s \gamma_{fs}\beta_{gs}}{\sum_s \gamma_{fs}\beta_{fs}}\right)}_{\equiv \kappa_{fg}(\gamma_f, \beta)} \pi_g.$$

- κ_{fg} is a profit weight: a dollar of profit at g is worth κ_{fg} dollars to f. It is the $\mathcal{H}(\kappa)$ of the Bertrand FOC, and is not specific to a particular strategic game.
- β (cash-flow rights) is often directly observable; γ embeds a model of corporate governance. The literature assumes $\gamma = \beta$ ("proportional control").

Reasons for Deviations from Multiproduct Bertrand

Biased estimates of own and cross price derivatives: For anything to work, you need correct estimates of Δ. My prior is most papers underestimate diversion ratios for close substitutes.

Vertical Relationships: Who sets supermarket prices? Just the retailer? Just the manufacturer? Some combination of both? Retailers tend to soften downstream price competition.

Faulty Timing Assumptions: Bertrand is a simultaneous move pricing game. Lots of alternatives (Stackelberg leader-follower, Edgeworth cycles, etc.).

Dynamics and Dynamic Pricing: Forward looking firms or consumers might not set static Nash prices. [e.g. Temporary Sales, Switching Costs, Network Effects, etc.]

Unmodeled Supergame: Maybe firms are legally tacitly colluding, higher prices might be about what firms believe will happen in a price war.

Where the Instruments Come From

In the notation of the demand lectures (characteristics x, demand shifters v, cost shifters w, demographics y):

$$\begin{aligned} \text{demand:} \quad & \delta_{jt}(\mathbf{s}_t, \mathsf{y}_t; \theta_2) + \alpha p_{jt} = h_d(\mathsf{x}_{jt}, \mathsf{v}_{jt}) + \xi_{jt} \\ \text{supply, conduct } m: \quad & p_{jt} - \eta^m_{jt}(\mathbf{p}_t, \mathbf{s}_t; \theta_2) = h_s(\mathsf{x}_{jt}, \mathsf{w}_{jt}) + \omega^m_{jt} \end{aligned}$$

- **Demand first.** Estimate θ_2 from the demand CMR $\mathbb{E}[\xi_{jt} \mid \mathsf{x}_t, \mathsf{v}_t, \mathsf{w}_t, \mathsf{y}_t] = 0$, then compute η^m for every m. The demand CMR is the same for every conduct model: stacking it with supply in one GMM (BLP 1995) only adds common overidentifying restrictions. Conduct enters through the supply CMR alone.
- For any m some $\mathbf{mc}$ rationalizes the data (J equations, J unknowns). The supply CMR has content only through exclusion: variables that move markups but not own cost (Berry Haile 2014).
- Candidates: demand shifters v and demographics y (they rotate demand); rivals' characteristics x_{-j} and rivals' cost shifters w_{-j} (they move rivals' prices, hence my markup).

From here on: x is every included cost control (x, w, fixed effects), z^e the excluded instruments, and $\mathbf{z} = (z^e, x)$. Demand, and so each η^m, is taken as known.

Classic approaches

Three Classic Ways to Use the Supply Restriction

Each conduct m implies a marginal cost $MC^m = p - \eta^m$:

1. **Estimate** (Wald). Put the markup on the right and instrument for it with z^e: $p = h_s(x) + \lambda\,\eta^m + \omega$; test $\lambda = 1$ (Pakes 2017, with Wollmann's trucks). Or estimate κ directly. Often underpowered, and $\lambda \neq 1$ is hard to interpret.
2. **Fit one model** (Anderson–Rubin / GMM-J). Put the markup on the left, $MC^m = h_s(x) + \omega^m$, and test $\mathbb{E}[\omega^m \mid \mathbf{z}] = 0$. Bonnet and Dubois (2010) and Villas-Boas (2007) compare such fits across models (Rivers–Vuong, Cox), often with no excluded instruments at all.
3. **Menu** (likelihood ratio). Estimate demand and supply jointly under each m and compare fit (Bresnahan 1987, Nevo 1998).

All three use the same supply CMR. The question is which test to run on it when every model is wrong.

Table 1: Wollman & Pricing Equilibrium.

Taken from Pakes, 2017, *Journal of Industrial Economics.*

	Price	(S.E.)	Price	(S.E.)
Gross Weight	.36	(0.01)	.36	(.003)
Cab-over	.13	(0.01)	.13	(0.01)
Compact front	-.19	(0.04)	0.21	(0.03)
long cab	-.01	(0.04)	0.03	(0.03)
Wage	.08	(.003)	0.08	(.003)
$\widehat{Markup}$	.92	(0.31)	1.12	(0.22)
Time dummies?	No	n.r.	Yes	n.r.
R^2	0.86	n.r.	0.94	n.r.

Note. There are 1,777 observations; 16 firms over the period 1992-2012. S.E.=Standard error.

- $\lambda \approx 1$ for multiproduct oligopoly, and fit is good: $R^2 > 0.8$ (> 0.5 within vehicle).
- As a behavioral model, multiproduct Bertrand with estimated demand looks successful.
- But would an alternative $\mathcal{H}(\kappa)$ give $\lambda \neq 1$ or a lower R^2? How low before we "reject"?

Villas Boas (2007)

TABLE 10

p-Values for pairwise non-nested comparisons

	Alternative models						
H_0 model	1	2	3·1	3·2	4	5	6
1: Simple linear pricing	–	0·50	0·00	0·50	0·24	0·00	0·50
2: Hybrid	0·00	–	0·50	0·50	0·12	0·00	0·50
3.1: Zero wholesale margin	0·41	0·29	–	0·05	0·50	0·39	0·07
3.2: Zero retail margin	0·39	0·40	0·05	–	0·50	0·39	0·17
4: Wholesale collusion	0·49	0·48	0·50	0·50	–	0·48	0·50
5: Retail collusion	0·00	0·00	0·50	0·50	0·22	–	0·50
6: Monopolist	0·34	0·35	0·17	0·31	0·48	0·34	–
Chain size weighted							
1: Simple linear pricing	–	0·08	0·01	0·06	0·08	0·00	0·00
2: Hybrid	0·17	–	0·15	0·22	0·00	0·06	0·14
3.1: Zero wholesale margin	0·08	0·15	–	0·11	0·15	0·12	0·00
3.2: Zero retail margin	0·01	0·07	0·00	–	0·09	0·01	0·00
4: Wholesale collusion	0·00	0·05	0·04	0·09	–	0·00	0·02
5: Retail collusion	0·00	0·02	0·03	0·11	0·02	–	0·00
6: Monopolist	0·10	0·20	0·00	0·15	0·20	0·14	–

Notes: *p*-Values reported from non-nested, Cox-type (Smith, 1992) test statistics of the null model in a row being true against the specified alternative model in a column. Bottom part is a robustness check. It has the same format as above, but the non-nested comparisons are based on estimates for the case when the portion of the manufacturer's profit due to each retailer is weighted by the retailer's chain size.

Source: My calculations.

- Try out different models of price setting, compute η^m, compare fit.
- Models with higher markups also have more $MC^m < 0$: wrong conduct, or inelastic demand?
- The Cox test rejects in both directions: (4) rejects (5) and (5) rejects (4); likewise 3.1 and 3.2 in the top panel.

Which Test? Duarte, Magnolfi, Sølvsten, Sullivan (QE 2024)

- **Specification tests** of one model at a time (Wald on $\lambda = 1$, Anderson–Rubin / GMM-J, Cox): with enough data every model is rejected, because no conduct model is exactly right. Villas-Boas's two-way Cox rejections are the symptom.
- **Model selection** (Rivers–Vuong 2002): compare two models' criteria under the null that they are equally misspecified. Only RV behaves sensibly when both models are wrong, which is always.
- The catch: RV is degenerate when neither model is falsified by the instruments (their Proposition 3). DMSS define weak instruments for testing and give an effective-F diagnostic (`pyRVtest`).
- Their criterion uses K unconditional moments $\bar{\mathbf{m}}_m = \mathbb{E}[z^e \, \omega^m]$ in the 2SLS quadratic form $\bar{\mathbf{m}}_m' \, \mathbb{E}[z^e z^{e\prime}]^{-1} \bar{\mathbf{m}}_m$: the supply CMR contracted against K directions. BCS 1.0 uses one optimal direction; BCS 2.0 uses all of them.

Model Selection under Misspecification

The Supply CMR and the Null: Ranking Misspecified Models

Residualize each model's implied cost on the controls, then ask whether what is left is predictable from the excluded instruments:

$$\omega^m = MC^m - h_s^m(x), \quad h_s^m(x) \equiv \mathbb{E}[MC^m \mid x], \qquad \mu^m(\mathbf{z}) \equiv \mathbb{E}[\omega^m \mid \mathbf{z}], \qquad Q(m) \equiv \|\mu^m\|^2 = \mathbb{E}\big[\mu^m(\mathbf{z})^2\big].$$

- Correct conduct: $\mu^m \equiv 0$. Wrong conduct: $\mu^m \neq 0$, if the instruments are rich enough to tell (falsifiability; Berry Haile 2014). $\mathbb{E}[\omega^m \mid x] = 0$ holds by construction for every m, so all the content is in z^e.
- On real data every $Q(m) > 0$. The honest hypothesis is not "Bertrand is true" but equal fit (Vuong 1989, Rivers–Vuong 2002, Chen–Hong–Shum 2007): $H_0 : Q(m_1) = Q(m_2)$ against H_A: one model is closer to the CMR.
- A "test of conduct" is therefore a ranking; the model confidence set holds the models never significantly beaten, not the correct one.
- μ^m is a function. Contract it against one direction (BCS 1.0), K directions (DMSS), or estimate the norm (BCS 2.0). Notation: $\langle a, b \rangle = \mathbb{E}[a(\mathbf{z})\, b(\mathbf{z})]$, $\|a\|^2 = \langle a, a \rangle$.

Backus, Conlon, Sinkinson: the scalar test (BCS 1.0)

One Direction: The Scalar Moment and Rivers–Vuong

For any direction $g(\mathbf{z})$, iterated expectations collapse the CMR to a scalar:

$$\bar{m}_m = \langle \mu^m, g \rangle = \mathbb{E}[\omega^m \, g(\mathbf{z})], \qquad \widetilde{Q}(m) = \bar{m}_m^2, \qquad \Delta\widetilde{Q} = \bar{m}_1^2 - \bar{m}_2^2,$$

$$\mathsf{T}_{RV} = \frac{\sqrt{n}\,\Delta\widetilde{Q}}{\widehat{\sigma}} \xrightarrow{d} \mathcal{N}(0,1) \text{ under } \Delta\widetilde{Q} = 0.$$

- **Validity is measurability.** g must be a function of $\mathbf{z}$ alone. Then a correct model has $\bar{m}_m = 0$ in every direction.
- The raw markup difference fails: η^m depends on prices and shares, hence on ω, so $\mathbb{E}[\omega^m \, \Delta\eta] \neq 0$ even under true conduct. Its projection onto $\mathbf{z}$ is valid.
- $\mathsf{T}_{RV} < 0$ favors m_1. Which g? The one that best tells m_1 from m_2.

The Wrong Model Omits a Markup Difference

Write $\Delta\eta \equiv \eta^{m_1} - \eta^{m_2}$. Implied costs differ by the markup difference with the opposite sign, so

$$MC^{m_1} - MC^{m_2} = -\Delta\eta, \qquad h_s^{m_1} - h_s^{m_2} = -\mathbb{E}[\Delta\eta \mid x], \qquad \omega^{m_1} - \omega^{m_2} = -(\Delta\eta - \mathbb{E}[\Delta\eta \mid x]).$$

Project onto $\mathbf{z}$ (using $x \subseteq \mathbf{z}$) to get the discriminating direction:

$$d(\mathbf{z}) \equiv \mu^{m_1} - \mu^{m_2} = -(\mathbb{E}[\Delta\eta \mid \mathbf{z}] - \mathbb{E}[\Delta\eta \mid x]) \equiv -g(\mathbf{z}).$$

- If m_1 is true, $\omega^{m_2} = \omega^{m_1} + (\Delta\eta - \mathbb{E}[\Delta\eta \mid x])$: misspecifying conduct puts an omitted variable, the residualized markup difference, into the cost shock. The test looks for it.
- g is the predicted markup difference net of the controls: two regressions of $\Delta\eta$, one on $\mathbf{z}$ and one on x, differenced (Frisch–Waugh–Lovell / Robinson).
- Subtracting $\mathbb{E}[\Delta\eta \mid x]$ cancels exactly the difference between the two models' cost functions.

Why g Is the Optimal Single Direction

Collapse the difference in cost shocks against an arbitrary direction $b(\mathbf{z})$:

$$M(b) \equiv \mathbb{E}\big[(\omega^{m_1} - \omega^{m_2})\, b(\mathbf{z})\big] = \langle d, b\rangle, \qquad \text{s.e.}\big(\widehat{M}(b)\big) = \frac{\sigma\,\|b\|}{\sqrt{n}} \text{ when } \mathbb{V}(\omega^{m_1} - \omega^{m_2} \mid \mathbf{z}) = \sigma^2.$$

Choose b to maximize the signal-to-noise ratio. Cauchy–Schwarz gives $\langle d, b\rangle \le \|d\|\,\|b\|$, with equality iff $b \propto d$, so

$$\frac{\sqrt{n}\,\langle d, b\rangle}{\sigma\,\|b\|} \;\le\; \frac{\sqrt{n}\,\|d\|}{\sigma} = \frac{\sqrt{n}\,\|\mu^{m_1} - \mu^{m_2}\|}{\sigma}, \qquad \text{with equality iff } b \propto d \propto g.$$

- **Unique**: any direction not proportional to g has strictly less power to tell the two models apart.
- **Attains the full CMR**: the best single direction recovers the whole conditional-moment distance $\|\mu^{m_1} - \mu^{m_2}\|$.
- **Chamberlain (1987)**: on the line $\eta(\tau) = \tau\eta^{m_1} + (1-\tau)\eta^{m_2}$, $\mathbb{E}[\partial\omega(\tau)/\partial\tau \mid \mathbf{z}] = -g$, so g is the optimal instrument for τ (with heteroskedasticity, $g/\sigma^2(\mathbf{z})$).
- **Caveats**: optimal per pair (each pair has its own g), and direction is not strength: z^e must still predict $\Delta\eta$ beyond x.

Rivers–Vuong on g Reproduces the Full-CMR Ranking

Optimality was for the linear contrast $\bar{m}_1 - \bar{m}_2$; Rivers–Vuong uses squares. Both factor through the same inner product:

$$\Delta\widetilde{Q}(b) = (\bar{m}_1 - \bar{m}_2)(\bar{m}_1 + \bar{m}_2) = \underbrace{\langle d, b\rangle}_{\text{discrimination}} \underbrace{\langle r, b\rangle}_{\text{level}}, \qquad r \equiv \mu^{m_1} + \mu^{m_2}.$$

Normalize $\hat{g} = g/\|g\| = -d/\|d\|$. Then $\langle d, \hat{g}\rangle = -\|d\|$, $\langle r, \hat{g}\rangle = -\langle d, r\rangle/\|d\|$, and

$$\Delta\widetilde{Q}(\hat{g}) = \langle d, r\rangle = \langle \mu^{m_1} - \mu^{m_2},\, \mu^{m_1} + \mu^{m_2}\rangle = \|\mu^{m_1}\|^2 - \|\mu^{m_2}\|^2 = Q(m_1) - Q(m_2).$$

- In population the scalar test and the full CMR rank the models identically and share the null: nothing is lost by using one direction.
- Population only: in samples $\hat{g}$ is estimated, so bias, variance and first-stage rates differ from the full-CMR estimator (BCS 2.0).
- Maximizing the raw $\Delta\widetilde{Q}(b)$ instead would tilt b toward the bisector of d and r: a bigger statistic for a different null (equal projection, not equal fit).

When Does the Test Fail?

The RV variance degenerates exactly when neither model is falsified in the direction used, $\bar{m}_1 = \bar{m}_2 = 0$ (DMSS Proposition 3).

Two failures survive the optimal g; both are matters of identification, not of instrument choice:

1. **Equal conditional moments**, $d = 0$: g itself vanishes and there is nothing to rank. If $\mu^{m_1} \equiv \mu^{m_2} \equiv 0$ (both correct, or z^e irrelevant for both), this is the DMSS degeneracy. With a common misfit the limit is non-degenerate, but $\hat{g}$ is pure first-stage noise and the fixed-$\hat{g}$ SE is too small (9–15% rejections at nominal 5%).
2. **Proportional moments**, $\mu^{m_1} \propto \mu^{m_2}$. Canonical case: two constant-markup rules $p = a_m\, mc$ give $\mu^m = v/a_m$ with $v = \mathbb{E}[p \mid \mathbf{z}] - \mathbb{E}[p \mid x]$, so $\Delta\widetilde{Q} = \|v\|^2(a_1^{-2} - a_2^{-2})$ and the test consistently picks the larger markup: scale, not fit, and no instrument fixes it. Never compare two constant-markup conducts. Perfect competition is a straw man: "we reject PC" only says that prices project onto z^e.

Two more failures hit a generic direction b, which need not lie in $\text{span}\{\mu^{m_1}, \mu^{m_2}\}$:

3. $\langle d, b\rangle = 0$ with $d \neq 0$: $\bar{m}_1 = \bar{m}_2$, spurious equal fit, no power.
4. $b \perp \text{span}\{\mu^{m_1}, \mu^{m_2}\}$: $\bar{m}_1 = \bar{m}_2 = 0$, the original Vuong degeneracy.

Along $g \propto d \in \text{span}\{\mu^{m_1}, \mu^{m_2}\}$, (3) and (4) can happen only at $d = 0$. In samples a poor $\hat{g}$ (weak z^e) can bring them back.

Plugging In $\widehat{h}_s$ and $\widehat{g}$: Orthogonality and Cross-Fitting

Both nuisance functions are fit by machine learning and plugged in once, never refit in the bootstrap. Why is that safe? Perturb (h_s, g) by $(\delta h, \delta g)$:

$$\frac{d}{dt}\Big|_{t=0} \bar{m}_m = \underbrace{-\,\mathbb{E}[\delta h(x)\,\mathbb{E}[g \mid x]]}_{\text{cost direction}} + \underbrace{\mathbb{E}[\mu^m\,\delta g]}_{\text{instrument direction}}, \qquad \frac{d}{dt}\Big|_{t=0} \widetilde{Q}(m) = 2\,\bar{m}_m \cdot \frac{d}{dt}\Big|_{t=0} \bar{m}_m.$$

- **Cost direction: zero by construction.** $\mathbb{E}[g \mid x] = 0$ because we subtracted $\mathbb{E}[\Delta\eta \mid x]$ (Neyman orthogonality; Chernozhukov et al. 2018). The subtraction is not redundant even though $x \subseteq \mathbf{z}$: without it, error in $\widehat{h}_s$ enters the moment at first order.
- **Instrument direction: zero iff $\mu^m \equiv 0$.** A model that satisfies the CMR has its moment pinned at zero however badly g is estimated. With a true model on the menu, $\hat{g}$-error moves power, not size.
- **Both-wrong null** ($\bar{m}_1 = -\bar{m}_2 = -\|d\|^2/2$): a term $-\|d\|^2\langle r, \delta g\rangle$ survives along the level direction, and a fixed-$\hat{g}$ SE omits its variance: $\mathbb{V}(\mathsf{T}_{RV}) \to 1 + R_r^2\,(1 - R_{fs}^2)/R_{fs}^2$, with R_{fs}^2 the first-stage R^2 of $\Delta\eta$ on z^e net of x and R_r^2 the share of the level they explain. Report R_{fs}^2. A refit bootstrap restores the term, and the full-CMR criterion (no estimated direction) is immune.
- **Cross-fitting** over whole clusters (markets, never products within a market) removes own-observation overfitting. The remaining bias is a product of two nuisance errors, negligible once each converges faster than $n^{-1/4}$.

Inference for the Scalar Test

Aggregate to cluster totals $S_c^m = \sum_{i \in c} w_i \, \hat{\omega}_i^m \hat{g}_i$ (clusters are unions of whole markets, w_i observation weights), so $\bar{m}_m = \frac{1}{n} \sum_c S_c^m$. No method below refits a nuisance function; each reweights the cached totals.

- **Analytic (default).** Delta method with gradient $(2\bar{m}_1, -2\bar{m}_2)$ and a cluster-robust variance. Fails as $\bar{m}_m \to 0$, where $\hat{\sigma} \to 0$.
- **Multiplier on the influence function.** Rademacher weights e_c on the linearized contributions. Reproduces the analytic SE and its boundary problem (Kline–Santos 2012).
- **Score-level wild.** Perturb each moment before squaring, $\bar{m}_m^* = \bar{m}_m + \frac{1}{n} \sum_c e_c \tilde{S}_c^m$ with $\tilde{S}_c^m$ the centered totals, then square. The extra quadratic term takes over near the boundary, so the bootstrap law self-adapts with no pretest. Recenter, since squaring a mean biases it up: report $\Delta\widetilde{Q} - (\widehat{V}_1 - \widehat{V}_2)$, with $\widehat{V}_m$ the bootstrap variance of $\bar{m}_m^*$.
- **Wild cluster bootstrap-t.** Studentize each draw: the few-cluster refinement (Cameron–Gelbach–Miller 2008).

Algorithm (BCS 1.0)

For a pair (m_1, m_2); the menu test repeats over all pairs.

1. **Split.** Partition the C clusters into K folds; every observation inherits its cluster's fold, and every first stage reuses the split.
2. **Cost residuals, per model.** For each m and fold k, fit $\hat{h}_s^{m,(-k)}(x) = \hat{\mathbb{E}}[MC^m \mid x]$ on the other folds (fixed effects absorbed by a within-transform) and set $\hat{\omega}_i^m = MC_i^m - \hat{h}_s^{m,(-k)}(x_i)$. Computed once per model and reused in every pair.
3. **Instrument, per pair.** $\hat{g}_i = \hat{\mathbb{E}}^{(-k)}[\Delta\eta \mid \mathbf{z}]_i - \hat{\mathbb{E}}^{(-k)}[\Delta\eta \mid x]_i$. Regress $\Delta\eta$ itself: for a nonlinear learner, $\hat{\mathbb{E}}[\eta^{m_1} \mid \cdot] - \hat{\mathbb{E}}[\eta^{m_2} \mid \cdot] \neq \hat{\mathbb{E}}[\Delta\eta \mid \cdot]$.
4. **Moments.** Cluster totals S_c^m, moments $\bar{m}_m$, contrast $\Delta\widetilde{Q} = \bar{m}_1^2 - \bar{m}_2^2$.
5. **Inference.** T_{RV} with the analytic cluster SE; score-level wild or bootstrap-t near the boundary. $\mathsf{T}_{RV} < 0$ favors m_1.
6. **Menu.** Pairwise matrix of contrasts $\rightarrow$ model confidence set: the models never significantly beaten.

The full conditional-moment test (BCS 2.0)

The Full Conditional-Moment Test in One Slide (BCS 2.0)

Use all of $\mu^m(\mathbf{z}) = \mathbb{E}[\omega^m \mid \mathbf{z}]$, not one direction: $Q(m) = \|\mu^m\|^2$, $H_0 : Q(m_1) = Q(m_2)$. The scalar BCS moment, the DMSS 2SLS quadratic form, and this norm are one direction, K directions, and all directions of the same object.

$$\widehat{Q}(m) = \frac{1}{C(C-1)} \sum_{c \neq c'} (u_c^m)' u_{c'}^m, \qquad u_c^m = \widehat{\Sigma}^{-1/2} \sum_{i \in c} \widehat{\omega}_i^m \, b(z_i^e), \qquad b \text{ a series basis, controls partialled out.}$$

- **Cross-cluster U-statistic**: products of residuals from different markets are unbiased for $\|\mu^m\|^2$; the own-cluster terms carry the estimation noise that inflates any plug-in (the "positive shift"), so they are deleted.
- **Why bother**: the degenerate set is the same as the scalar test's (neither model falsified), so it buys no power there. What it buys is an instrument side with no estimated direction: where the scalar $\hat{g}$ is first-stage noise (equal conditional moments, the both-wrong null) the norm keeps nominal size. One cross-fit regression per model, nothing refit per pair.
- **The price**: under a correct model the criterion sits at its minimum and is not normal, so inference has two regimes (two slides on).

The Full-CMR Procedure: Which Regressions, and How the Criterion Is Formed

For a menu of models m; only step 2 is per model, nothing is refit per pair.

1. **Split.** Partition the C clusters (whole markets) into K folds; the same split serves every regression below.
2. **Cost residuals, per model.** Regress $MC^m = p - \eta^m$ on the controls x with the FE absorbed by a within-transform, out of fold: $\widehat{\omega}_i^m = MC_i^m - \widehat{h}_s^{m,(-k)}(x_i)$. The same residuals as the scalar test; the only nonparametric nuisance.
3. **Instrument basis.** Take a series basis $b(z^e) \in \mathbb{R}^{K_z}$ on the excluded instruments (a low-order polynomial). Regress every column on x, cross-fit, and keep the residuals $\tilde{b}(z^e)$: the controls are partialled out of the basis exactly as they are out of $\hat{g}$ (mean-centering is the constant-only case). Whiten with $\widehat{\Sigma} = \frac{1}{n}\sum_i \tilde{b}_i \tilde{b}_i'$.
4. **The CMR estimate is a regression.** Regress $\widehat{\omega}^m$ on $\tilde{b}$: $\widehat{\mu}^m = \tilde{b}'\,\widehat{\Sigma}^{-1}\,\frac{1}{n}\sum_i \tilde{b}_i \widehat{\omega}_i^m$, the fitted value. Its mean square, $\hat{\nu}_m'\hat{\nu}_m$ with $\hat{\nu}_m = \widehat{\Sigma}^{-1/2}\frac{1}{n}\sum_i \tilde{b}_i \widehat{\omega}_i^m$, is the 2SLS quadratic form of DMSS. It is biased upward by the estimation variance of $\widehat{\mu}^m$.
5. **Debias with cluster scores.** $G_c^m = \sum_{i \in c} w_i\, \widehat{\omega}_i^m \tilde{b}_i$, $u_c^m = \widehat{\Sigma}^{-1/2} G_c^m$, and $\widehat{Q}(m) = \frac{1}{C(C-1)}\left(\|\sum_c u_c^m\|^2 - \sum_c \|u_c^m\|^2\right)$: the cross-cluster products only. Dropping $c = c'$ drops $\mathbb{E}\|\widehat{\mu}^m - \mu^m\|^2$.
6. **Contrast and inference.** $\Delta\widehat{Q} = \widehat{Q}(m_1) - \widehat{Q}(m_2)$; per-cluster influence $\widehat{\psi}_c$ (leave-one-out $\hat{\nu}$, weight-matrix term); unified multiplier bootstrap on the cached u_c^m; p-value from the bootstrap law (next slide).
7. **Menu.** Pairwise matrix of contrasts $\rightarrow$ model confidence set.

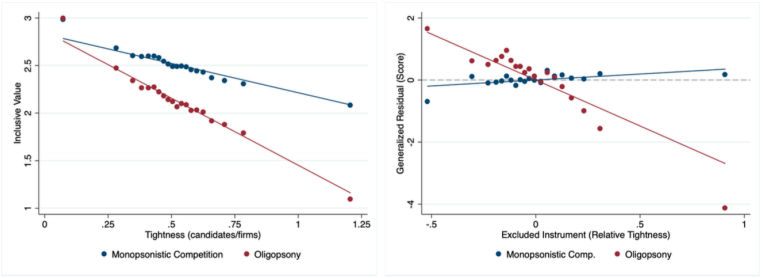
Inference in One Slide: Two Regimes, One Bootstrap

$$\Delta\widehat{Q} - (Q(m_1) - Q(m_2)) = \underbrace{\frac{1}{C}\sum_c \psi_c}_{\sqrt{C}\text{-normal if } \sigma^2_{\text{lin}} > 0} + \underbrace{\frac{1}{C(C-1)}\sum_{c \neq c'} \Psi(c, c')}_{\text{rate-}C \text{ weighted } \chi^2 \text{ if } \mu^{m_1} \equiv \mu^{m_2} \equiv 0}$$

- **Non-degenerate**: at least one model violates the CMR on the span of z^e; cluster-robust RV is fine. **Degenerate**: neither does (both correct, or weak instruments for testing); the linear term vanishes, $C\widehat{Q} \to \sum_r v_r(\chi^2_{1,r} - 1)$, mean zero and right-skewed, so a normal cutoff under-covers the upper tail. The standard error differs by $\sqrt{C}$ between regimes and weak-instrument comparisons sit near the boundary. Because $\widehat{\Sigma}$ is estimated, ψ_c carries a weight-matrix term: per observation the influence of $Q(m)$ is $2v\omega^m - v^2 - Q(m)$ with v the fitted moment (DMSS eq. 8); dropping v^2 overstates the variance.
- **One multiplier bootstrap** on both terms, $\Delta\widehat{Q}^* - \Delta\widehat{Q} = \frac{1}{C}\sum_c e_c\widehat{\psi}_c + \frac{1}{C(C-1)}\sum_{c \neq c'} e_c e_{c'}\widehat{\Psi}(c, c')$, one Rademacher weight per cluster shared across terms and models. It nests the linear RV and the score-wild bootstrap and interpolates between them with no pretest and no switch; the p-value comes from the bootstrap law, not a normal table; nothing is refit.
- **Diagnostics**: $\widehat{F} = \max_m C\widehat{Q}(m)$ over the pair (near degeneracy only when both criteria are near zero) and the first-stage fit of $\Delta\eta$ on z^e (is there anything to rank?), both descriptive. **Open**: uniform validity across the boundary (the Shi 2015 analogue), demand-estimation uncertainty, a Hansen model confidence set.

Examples

Figure 5: Vuong Test



(a) First Stage **(b)** Visualizing the Vuong Test

Note: Panel (a) plots the "first stage" relationship between the model-implied inclusive values Λ_i and Λ_i^{-j} and the instrumental variable t_{ij}, conditional on firm covariates z_j and candidate covariates x_i and two-week period dummies. Panel (b) plots the relationship between generalized residuals and the excluded instrument for the non-predictive monopsonistic competition and oligopsony models. Under proper specification, the correlation of the generalized residuals and the excluded instrument should be zero (the dashed line). The larger the deviation from zero, the greater the degree of mis-specification of the model.

- Offered wages for online job platform
- Compares Monopsony vs. oligopolistic competition vs. perfect comp.
- Compares tailored offers vs. not (price discrimination).

Scuderi JMP 2022: Models of Labor Supply

- Firms ignore competitors (Monopsony)
- Firms offer wages independent of candidate characteristics (experience, demographics).
- Firms are definitely NOT paying MPL.

Table 4: Non-Nested Model Comparison Tests (Rivers and Vuong, 2002)

Model	(1) Monopsonistic Comp.	(2) Monopsonistic Comp.	(3) Oligopsony	(4) Oligopsony
	Not Predictive	Type Predictive	Not Predictive	Type Predictive
Perfect Competition	-54.84	-54.40	-39.92	-39.91
Monopsonistic, Not Predictive	–	7.83	3.98	2.69
Monopsonistic, Type Predictive		–	2.77	1.54
Oligopsony, Not Predictive			–	-3.67
Oligopsony, Type Predictive				–

Note: This table reports test statistics from the Rivers and Vuong (2002) non-nested model comparison procedure. Positive values imply the row model is preferred to the column model. Under the null of model equivalence, the test statistics are asymptotically normal with mean zero and unit variance.

specification. In the figure, the generalized residuals for the monopsonsitic competition alternative are closely aligned with the x-axis, while the generalized residuals for the oligopsony alternative are strongly negatively related to tightness.

Our tests therefore suggest that models of firm behavior in which firms both ignore strategic interactions in wage setting and do not tailor wage offers to candidates on the basis of predictable preference variation are closer approximations to firms' true bidding behavior on the platform than are models in which firms act strategically and tailor offers. In Appendix F, we report the testing results using the original Vuong (1989) likelihood comparison test, which yield qualitatively identical model comparisons. In the following analysis, we adopt the not-predictive monopsonostic competition model as our preferred model of conduct.

Starc Wollmann: Generic Pharma Cartel + Entry

Do cartels encourage entry with high prices? Starc and Wollmann (2022) apply the BCS test to the generic-drug cartel; the menu is competitive vs. joint-profit-maximizing pricing before and after cartel formation, with non-members best responding.

> *in 2013, Teva Pharmaceuticals, the largest generic firm, hired NP, a marketing executive with especially strong industry relationships, and tasked her with "price increase implementation."1 Over an 18-month period, industry participants exchanged thousands of calls and texts—alongside countless LinkedIn, Facebook, and WhatsApp messages and face-to-face conversations—with contacts at rival firms to coordinate the increases (Complaint, page 322).2 Following this period, prescription drug expenditures by governments, private insurers, and individuals rose sharply by billions of dollars.*

Starc Wollmann: Testing Results (prices rise with the cartel, less in markets that invite entry)

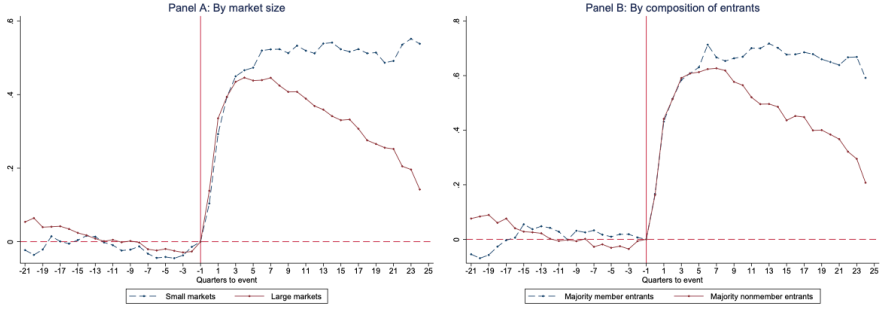
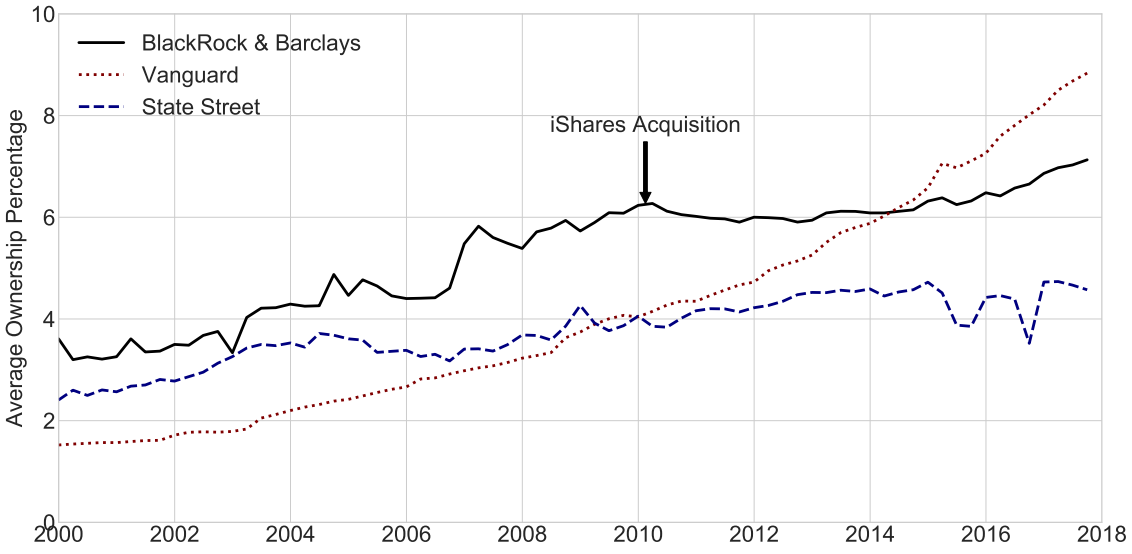


Figure III: *Prices after cartel formation*

In this figure, we plot estimates of β^{τ}, which are derived from equation 1, against x^{τ}, which represents event time in quarters. For Panel A, we proxy for market size by computing the number of prescriptions filled in the period just prior to NP joining Teva, and we distinguish between large and small cartelized markets based on how this figure relates to the median value. For Panel B, we calculate the fraction of entrants that arrive as a result of ANDAs filed after NP joins Teva and identify markets in which a majority of those firms are members and nonmembers, respectively.

Application: Common Ownership (Backus Conlon Sinkinson)

Rise of the Big Three (S&P 500 components, BCS AEJ:Micro 2021)



The Reduced-Form Literature (and its Critics)

Symmetric homogeneous-good Cournot with profit weights gives a share-weighted average markup (with market demand elasticity ϵ):

$$\sum_f s_f \cdot \frac{p - mc_f}{p} = \frac{1}{\epsilon} \underbrace{\mathbf{s}' \mathcal{H}(\kappa) \mathbf{s}}_{\equiv MHHI}, \qquad MHHI\Delta \equiv MHHI - \underbrace{\mathbf{s} \cdot \mathbf{s}}_{HHI}$$

Some regress

$$\log\left(p_{rjt}\right) = \beta_1 \cdot MHHI\Delta_{rt} + \beta_2 \cdot HHI_{rt} + \theta \cdot X_{rjt} + \alpha_t + \nu_{rj} + u_{rjt}$$

- Azar, Schmalz, Tecu (JF 2018): airline routes with more common ownership have 3–7% higher fares.
- But $MHHI\Delta$ is a function of equilibrium market shares – endogenous! (O'Brien Waehrer 2017). Price-concentration regressions all over again.
- Dennis, Gerardi, Schnitzler (JF 2022): not robust (driven by the share component); Kennedy, O'Brien, Song, Waehrer (2017): structural version finds nothing.
- Better idea: treat CO vs. own-profit maximization as a conduct testing problem.

The Menu Is an Internalization Ladder

In cereal the candidate conducts differ in how much of the cross-price structure the firm prices on:

$$\underbrace{\text{single-product}}_{\text{own elasticities only}} \prec \underbrace{\text{own-profit}}_{\text{Nash–Bertrand, own firm}} \prec \underbrace{\text{common ownership } \kappa}_{\text{own firm + profit weights on rivals}} \prec \underbrace{\text{monopoly}}_{\text{everything}}$$

- All of them, single-product included, price above marginal cost. The test ranks models by the recapture they internalize, not by whether market power exists.
- So "single-product fits the moment best" is a statement about cross-price internalization, not a rejection of market power, and not a statement that firms ignore their sister brands in any literal sense.
- Perfect competition is on the menu as a straw man only (the constant-markup degeneracy). Cournot vs. Bertrand would be a different ladder.
- This is the substantive content of the criterion $Q(m) = \|\mathbb{E}[\omega^m \mid \mathbf{z}]\|^2$ here: it orders the ladder; it does not estimate a markup level. The markup level comes from demand.

Near-Collinear Markups Lower Power; Instrument Strength Sets the Regime

- Across the menu the implied markups are ≥ 0.92 correlated, so $\Delta\eta = \eta^{m_1} - \eta^{m_2}$ is small relative to η^m. That does not put the test near the degeneracy boundary: the scale of $\Delta\eta$ cancels from the studentized statistic. It lowers power, because the misfit a wrong model adds is small relative to the noise in the cost residual.
- What the test needs is that the excluded z predict $\Delta\eta$ net of the controls: instrument strength for conduct, a different object from strength for demand. An instrument can move prices and shares beautifully and project onto $\Delta\eta$ not at all.
- Monte Carlo with one model correct: rejection $1.00 \to 0.05$ as $\text{corr}(\eta^{m_1}, \eta^{m_2})$ rises from 0.97 to 0.999. That is power falling as the wrong model's violation shrinks, with both moments going to zero, not a degeneracy caused by collinear markups.
- Practical implication: before testing, regress $\Delta\eta$ on the instruments (net of controls) and report the fit, the effective first-stage F. If nothing explains the markup difference beyond the cost shifters, no test, scalar or norm, will rank the pair. This is the DMSS weak-instruments-for-testing diagnostic in plain clothes, and for the scalar test the same R^2_{fs} sets how far its fixed-$\hat{g}$ SE is off.

Validity Is Orthogonality to the Cost Shock, Not to Price

A conduct instrument must be excluded from ω; it must correlate with price, since a useful instrument moves markups. Choosing price-orthogonal instruments manufactures weak ones. The cereal instrument groups, after partialling the controls x (FE included) out:

Group	What it is	Strength for conduct
`opt_iv`	demand-side Chamberlain optimal IV, $\mathbb{E}[\partial\xi/\partial\theta \mid \mathbf{z}]$ at a flexible $\widehat{\mathbb{E}}[p \mid \mathbf{z}]$	strong (effective $F \approx 50$–60)
`demog`	chain-level income and demographic moments	moderate ($F \approx 10$–17); mostly absorbed by the FE
`opc`	price of a grain the product does *not* use (a rival's cost shifter)	valid; $\approx 10\%$ of its variance survives the FE ($F \approx 5$–12)
`diff`	BLP / Gandhi–Houde differentiation instruments	relevant ($F \approx 30$–60)

- The optimal IV is not circular: demand was estimated without a supply side, so $\widehat{\mathbb{E}}[p \mid \mathbf{z}]$ is a reduced-form fit, not an equilibrium price under an assumed conduct. Its near-perfect span of expected prices is the source of its power.
- The BLP differentiation instruments are built for demand. Their strength for conduct is a separate empirical question, answered by the first-stage F for $\Delta\eta$ and not by their strength for demand.
- Partial the controls out of the instrument basis (FWL), as the scalar test partials $\hat{g}$; otherwise control overlap inflates the Gram matrix and groups are not comparable. Partialling sharpens valid strong instruments and exposes FE-collinear ones as weak.

Fixed Effects, Weighting, and the Specification of h_s

- Fixed effects in h_s absorb cost heterogeneity that would otherwise read as moment violation, and they absorb each instrument's FE-collinear variation: the chain-level demographics and the product-level grain prices lose most of theirs. Product fixed effects matter most.
- Example: across a 20-cell sweep (5 FE sets $\times$ 2 weightings $\times$ 2 instrument groups) the model confidence set is $\{\text{single}\}$ in every cell with product FE; without product FE and without weights the collusive models survive under the optimal IV.
- Servings weighting roughly halves the criterion magnitudes. Additive versus interacted FE can flip a pairwise decision; the full product$+$year$+$chain set is invariant.
- Lesson: the specification of h_s (what is a cost control, which FE, how flexible, weighted or not) is as much a part of the identification argument as the instruments. Report it, every time.

Scalar Test Results (BCS 1.0): Rivers–Vuong T, $N(0,1)$

	Others' Cost	Demographics	BLP Inst.	Dmd. Opt. Inst.
Own Profit Max vs.	Panel 1: $A(\mathbf{z}_t) = \mathbf{z}_t$, linear $h_s(\cdot)$			
Single Product	1.6489	1.1116	0.9977	1.6432
Common Ownership	-3.8928	-1.1957	0.5044	-1.0329
Double Marginalization	1.4435	0.9892	-0.0427	5.5684
Common Ownership + Double Marginalization	-0.1919	0.6815	0.1404	5.4688
Perfect Competition	1.1730	0.4171	0.7364	3.9589
Monopolist	-1.4097	-1.0680	-0.4523	-1.0908
Own Profit Max vs.	Panel 2: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12}\|\mathbf{z}_t]$, linear $h_s(\cdot)$ and $g(\cdot)$			
Single Product	1.4264	0.5795	0.6662	0.7516
Common Ownership	-2.3044	-0.5105	-0.0384	-1.4297
Double Marginalization	0.8644	0.4421	-0.5311	2.9800
Common Ownership + Double Marginalization	-0.9382	-0.2389	-0.3684	0.2460
Perfect Competition	0.7164	0.6135	-0.1080	1.8776
Monopolist	-0.8577	-0.4002	-0.3868	-1.1097
Own Profit Max vs.	Panel 3: $A(\mathbf{z}_t) = \mathbb{E}[\Delta\eta^{12}\|\mathbf{z}_t]$, random forest $h_s(\cdot)$ and $g(\cdot)$			
Single Product	5.0725	5.3347	5.2702	5.5758
Common Ownership	-3.9823	-3.6775	-4.0786	-4.3977
Double Marginalization	-6.3295	-9.9033	-6.5311	-7.6302
Common Ownership + Double Marginalization	-6.6568	-6.8735	-6.6624	-7.5578
Perfect Competition	-5.5005	-7.7749	-6.4083	-7.7653
Monopolist	-3.7240	-4.4602	-3.6134	-3.8959

Example results: RTE cereal (NielsenIQ, 2007–2017), BLP demand with micro moments; the menu is single-product, own-profit, common ownership κ, and monopoly pricing. $\mathsf{T} < 0$ favors own-profit. Every entry is conditional on the estimated demand system that produced the markups.

Full Conditional-Moment Results (BCS 2.0): Own-Profit Against Each Alternative

Studentized contrast $\Delta\widehat{Q}$ of own-profit against the row model, unified-bootstrap p-value in parentheses; $Z < 0$ favors own-profit. Product+year+chain FE, servings weights, boosting, controls partialled out of the instrument basis.

own-profit vs.	`opt_iv` Z (p)	`demog` Z (p)
single-product	$+5.22\,(< .001)$	$+4.18\,(< .001)$
common ownership κ	$-3.11\,(.005)$	$-4.09\,(< .001)$
common ownership κ_{lag}	$-2.57\,(.012)$	$-3.96\,(< .001)$
common ownership κ_{ma}	$-2.46\,(.024)$	$-4.28\,(< .001)$
monopoly (big 4)	$-3.82\,(< .001)$	$-4.82\,(< .001)$

- **Same ladder as the scalar test**: single-product beats own-profit, and own-profit beats common ownership and monopoly, under both instrument groups with power.
- The rivals' cost shifters (`opc`) have no power for any pair; the differentiation instruments (`diff`) separate single-product from own-profit ($+4.91$) and nothing else. Strength for conduct, not validity, is what limits them.
- Every $Q_m > 0$: this is "rank the misspecified models," conditional on the demand-side instruments being excluded from the true cost shock, and on the estimated demand system.

The Single-Product Puzzle

- Single-product pricing (each brand ignores its sister brands) fits the moment condition better than own-profit maximization. Robust across instruments, samples, fixed effects, weighting, learners, and both criteria; look back at the scalar-test table.
- It is not "no market power": single-product markups are $1/(\alpha(1-\sigma_j))$-sized, comfortably above cost. It says firms appear to internalize less cross-brand recapture than Bertrand with the estimated demand implies.
- Candidate explanations, none settled:
  1. Demand misspecification: the cross-price elasticities (recapture) are mismeasured, so the "correct" internalization looks wrong through the lens of this demand system. Recall the micro-data week: substitution patterns are the thing BLP gets least right.
  2. Rule-of-thumb pricing: brand managers price their own brand; the corporate parent does not reprice Cheerios when Lucky Charms moves.
  3. Another instrument or moment may reorder single vs. own-profit; nothing we have tried does.
- Methodological point: a result like this only becomes visible once the test is honest about ranking misspecified models. A specification test would have rejected everything and said nothing.

Stepping Back: What About the Mechanism?

- In RTE cereal, the evidence favors own-profit maximization over common ownership pricing, on BCS 1.0 and BCS 2.0 alike.
- In some sense this is a frictionless model
  - Managers do exactly what investment managers want
  - Investment managers seek to maximize portfolio value
- Some attempts at micro foundations are questionable
  - Anton, Ederer, Gine, Schmalz: toy model has manager choose effort to reduce mc.
  - Weaker managerial incentives: higher mc and higher p
  - Some correlation that executives at commonly owned firms have performance less correlated with profitability.
- In some sense CO is what we would see absent agency problems, so where are they?

Thanks!